

Short Technical Update: From Baseline MLP to MLP Run 005

Hexapod RL Project Update

February 13, 2026

1. Objective

The goal was to improve locomotion realism for a servo-driven hexapod while preserving forward motion. In practical terms, we targeted:

- slower and cleaner leg motion (less wobble/noise),
- clearer lift–swing–land behavior,
- stable path-following, and
- retained (or improved) displacement in the environment.

2. Baseline MLP Characteristics

The original baseline run (`runs/mlp_baseline`) used a fast phase and imitation rhythm (0.9 Hz), no contact observation, and no explicit smoothness/jerk/stance penalties. The reward strongly emphasized uncapped positive forward velocity (`forward_velocity_weight = 50.0`) with relatively light regularization.

Observed behavior from baseline videos and metrics:

- high-frequency leg activity and visible wobble,
- occasional large spikes in velocity,
- less servo-plausible command profile.

3. Main Changes Introduced Before Run 005

Relative to baseline MLP, the following changes were applied and then consolidated in `runs/mlp_run_005/config.yaml`:

3.1 Motion Timing and Servo Plausibility

- Phase and imitation frequencies reduced from 0.9 Hz to 0.45 Hz.
- Action smoothing enabled: `action_smooth_alpha = 0.6`.
- Step-wise action-rate cap enabled: `action_rate_limit = 0.15`.

3.2 Contact-Aware Gait Structuring

- Contact observation enabled (`obs_include_contact = true`).
- New stance penalties used to reduce motion while feet are planted:
 - `stance_stability_penalty_weight = 0.2`,
 - `stance_action_penalty_weight = 0.35`.

3.3 Smoother Control and Anti-Wiggle Regularization

- `smoothness_reward_weight = 2.5`,
- `jerk_penalty_weight = 0.35`,
- `joint_velocity_penalty_weight = 0.04`.

3.4 More Deliberate Step Geometry and Locomotion Target

- Relaxed motion amplitudes:
 - `action_scale_abd_deg`: 35 $\rightarrow$ 42,
 - `action_scale_flex_deg`: 60 $\rightarrow$ 72,
 - `max_flexion_angle`: 70 $\rightarrow$ 88.
- Switched from direct forward-velocity reward to target-speed shaping:
 - `forward_velocity_weight` = 0,
 - `target_forward_velocity` = 0.09,
 - `target_velocity_weight` = 28.0.

4. Quantitative Snapshot (Baseline vs Run 005)

Metric (evaluation summary)	Baseline MLP	MLP Run 005
Mean reward	2817.78	21962.45
Reward std. dev.	3208.06	71.16
Mean forward velocity (m/s)	0.01266	0.01881
Max forward velocity (m/s)	1.0795	0.1750
Mean torso height (m)	0.09976	0.08774
Mean energy proxy ($\sum a^2$)	3.8585	1.0746
Mean episode length (steps)	1000	1000

Interpretation:

- Run 005 is far more regular and stable (very low reward variance).
- Mean forward speed improved versus baseline, while extreme velocity spikes dropped substantially.
- Energy proxy decreased strongly, consistent with smoother servo-friendly command behavior.

5. Visualization/Analysis Improvements Added

To improve interpretability of locomotion relative to the environment, evaluation video tooling was upgraded:

- Per-frame overlay now includes step index, (x, y) position, forward distance, and forward velocity.
- For each saved episode video, a companion trajectory plot (`_trajectory.png`) is generated showing:
 - XY path in world coordinates (start/end markers),
 - forward distance and velocity vs time.

These additions make it easier to diagnose whether gait quality improvements still preserve useful displacement.

6. Conclusion

Compared with the baseline MLP, Run 005 applies a contact-aware, servo-realistic control strategy that meaningfully improves gait realism and smoothness while maintaining locomotion. The resulting behavior is closer to an insect-like tripod-style pattern and avoids the unrealistic high-frequency wobble seen in the baseline.

In short: **the project successfully moved from “fast but noisy” baseline locomotion to “cleaner and physically plausible” locomotion with retained movement performance.**